Session1: Review of Fixed-Effects Regression

Time to know more

Instructor: Mohammad Arashi (FUM, UP)

دانشگاه فردوسی مشهد

Ferdowsi University of Mashhad

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Outline

Multiple Regression Model-Fixed effects

Consider the set of observations $\{(\boldsymbol{x}_i^\top, y_i)\}_{i=1}^n$ from the **mean regression** model $\mathrm{E}(y|\boldsymbol{x}) = \boldsymbol{x}^\top\boldsymbol{\beta}$ with $\boldsymbol{x}_i \in \mathbb{R}^p$, regression vector $\boldsymbol{\beta} = ({}^1\beta_1, \ldots, \beta_p)^\top$. This relation can become more informative (as we want to see) by putting all we did not consider in the modeling in an error term and equivalently show the model as

$$\begin{aligned} & y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i = \boldsymbol{x}_i^\top\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad i = 1, 2, \ldots, n \\ \equiv \quad & \boldsymbol{y} = \boldsymbol{X}\beta + \boldsymbol{\epsilon}, \end{aligned} \tag{1.1}$$

with

$$\boldsymbol{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \boldsymbol{X} = \begin{pmatrix} \boldsymbol{x}_1^\top \\ \vdots \\ \boldsymbol{x}_n^\top \end{pmatrix} \sim n \times p, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

[1]We eliminated the intercept parameter by centering the observations

We do not take specific distributional assumption at this stage and only assume $\mathrm{E}(\boldsymbol{\epsilon}) = \mathbf{0}$ (why?) and $\mathrm{E}(\boldsymbol{\epsilon}^\top \boldsymbol{\epsilon}) = \sigma^2 \boldsymbol{I}_n$, $\sigma^2 > 0$. Then, the ordinary least squares (OLS) estimate of $\boldsymbol{\beta}$ is obtained as

$$\hat{\boldsymbol{\beta}} \quad = \quad \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \{ S(\boldsymbol{\beta}) = \boldsymbol{\epsilon}^\top \boldsymbol{\epsilon} \}, \quad \boldsymbol{\epsilon} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta} \tag{1.2}$$

The system of normal equations can be obtained from $\frac{\partial}{\partial \boldsymbol{\beta}} S(\boldsymbol{\beta}) = \mathbf{0}$ as

$$\boldsymbol{C}\boldsymbol{\beta} = \boldsymbol{X}^\top \boldsymbol{y}, \quad \boldsymbol{C} = \boldsymbol{X}^\top \boldsymbol{X}. \tag{1.3}$$

If $\boldsymbol{C}$ is invertible, then we have the estimates as

$$\hat{\boldsymbol{\beta}} = \boldsymbol{C}^{-1} \boldsymbol{X}^\top \boldsymbol{y}. \tag{1.4}$$

Linear algebra play

Is $\boldsymbol{C}$ always invertible?

If $\boldsymbol{C}$ is full rank then, it is invertible[2] (nonsingular). If $n > p$, then we will assume $\text{rank}(\boldsymbol{X}) = p$ (full column rank) and therefore $\text{rank}(\boldsymbol{C}) = \text{rank}(\boldsymbol{X}) = p$. In this case, the solution to the optimization problem $\min S(\beta)$ is given by

$$\hat{\beta} = \boldsymbol{C}^{-1}\boldsymbol{X}^\top \boldsymbol{y}.$$

condition number: $\kappa(\boldsymbol{C}) = \frac{\lambda_{\max}(\boldsymbol{C})}{\lambda_{\min}(\boldsymbol{C})}$.

The matrix $\boldsymbol{C}$ is ill-conditioned if the condition number is too large (and singular[3] if it is infinite). How can it be singular?. Simply, if one or some of the eigenvalues are zero then the condition number is infinite.

[2] Remember $\text{rank}(\boldsymbol{C}) \leq \min(\text{rank}(\boldsymbol{X}), \text{rank}(\boldsymbol{X}^\top)) = \min(n, p)$

[3] non-invertible

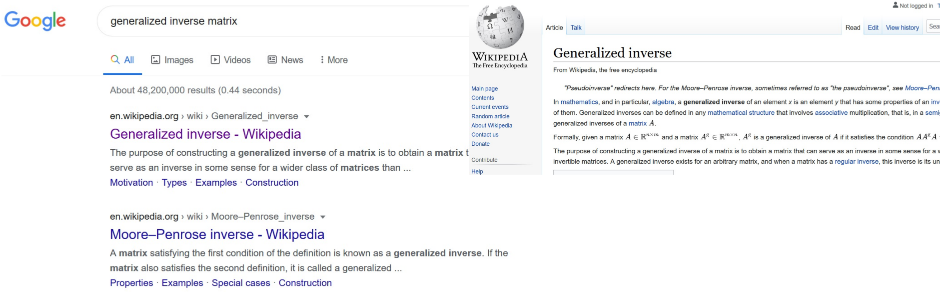
Remedy (R)

If $p \geq n$ [4] then, $\boldsymbol{C}$ is singular. In general, how can we find the solution (1.2) if $\boldsymbol{C}$ is not invertible?

- R1 : Use the generalized inverse of $\boldsymbol{C}$ (with care).
- R2 : Use the ridge trick.
- R3 : Perform variable selection and use the oracle estimate (with care).

[4] Termed as high-dimensional case

Moore Penrose generalized inverse



Use $\hat{\beta} = (\boldsymbol{X}^\top \boldsymbol{X})^{+} \boldsymbol{X}^\top \boldsymbol{y}$

Toy simulation in R

Lab1:
n = 50
p = 60
X = matrix(0, n, p) ...
We use pseudo or Moore-Penrose generalized inverse of $\boldsymbol{X}^\top \boldsymbol{X}$
library("pracma")
See also ginv Inference is not easy!

No suppose $\boldsymbol{C}$ is ill-conditioned. Then the eigenvalues are given by $\lambda_1, \cdots, \lambda_p$. Obviously if some of the eigenvalues are zero, adding a positive constant $k > 0$ to all, will give non-zero eigenvalues $\lambda_1 + k, \ldots, \lambda_p + k$. These latter eigenvalues belong to the matrix $\boldsymbol{C} + k\boldsymbol{I}_p$. This is called the ridge trick[5] (refer to Assignment Q2)
Then, the ridge regression (RR) estimator is given by

$$\begin{aligned} \hat{\beta}(k) &= \boldsymbol{C}(k)^{-1}\boldsymbol{X}^\top \boldsymbol{y}, \quad \boldsymbol{C}(k) = (\boldsymbol{C} + k\boldsymbol{I}_p),\ k > 0 \\ &= (\boldsymbol{C} + k\boldsymbol{I}_p)^{-1}\boldsymbol{X}^\top \boldsymbol{y}. \end{aligned} \tag{3.1}$$

The parameter $k \in \mathbb{R}^+$ is called the ridge parameter or shrinkage parameter.

[5] Refer to: Saleh, A.K.Md.Ehsanes, Arashi, M. and Kibria, B.M.Golam (2019) Theory of Ridge Regression Estimation with Applications, John Wiley, New York.

Let's come back again to small eigenvalues...
What is the problem of having small eigenvalues?
We need to work with canonical form to understand the problem.

Spectral decomposition

Since $\boldsymbol{C} : p \times p$ is symmetric we perform a canonical analysis[6] on its structure. Let $\{(\lambda_i, \boldsymbol{u}_i)\}_{i=1}^{p}$ be the set of (aka eigenvalues,eigenvectors) of $\boldsymbol{C}$, then we can decompose $\boldsymbol{C}$ as

$$\boldsymbol{C} = \underbrace{\begin{pmatrix} \uparrow & \uparrow & \cdots & \uparrow \\ \gamma_1 & \gamma_2 & \cdots & \gamma_p \\ \downarrow & \downarrow & \cdots & \downarrow \end{pmatrix}}_{\text{Orthogonal matrix } \boldsymbol{\Gamma}} \overbrace{\begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_p \end{pmatrix}}^{\text{Diagonal matrix } \boldsymbol{\Lambda}} \underbrace{\begin{pmatrix} \leftarrow & \gamma_1^\top & \rightarrow \\ \leftarrow & \gamma_2^\top & \rightarrow \\ \leftarrow & \vdots & \rightarrow \\ \leftarrow & \gamma_p^\top & \rightarrow \end{pmatrix}}_{\boldsymbol{\Gamma}^\top}$$

$$= \sum_{j=1}^{p} \lambda_j \gamma \gamma^\top. \tag{3.2}$$

[6] In linear algebra, eigen decomposition or spectral decomposition is the factorization of a matrix into a canonical form, where the matrix is represented in terms of its eigenvalues and eigenvectors.

Canonical regression model

Recall the model $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ and use the orthogonal matrix $\boldsymbol{\Gamma}$ to write

$$\begin{aligned} \boldsymbol{y} &= \boldsymbol{X}\boldsymbol{\Gamma}\boldsymbol{\Gamma}^\top\boldsymbol{\beta} + \boldsymbol{\epsilon} \\ &= \boldsymbol{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \quad \boldsymbol{Z} = \boldsymbol{X}\boldsymbol{\Gamma},\ \boldsymbol{\theta} = \boldsymbol{\Gamma}^\top\boldsymbol{\beta}. \end{aligned} \tag{3.3}$$

Then, the OLS estimate of $\boldsymbol{\theta}$ is given by

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= (\boldsymbol{Z}^\top\boldsymbol{Z})^{-1}\boldsymbol{Z}^\top\boldsymbol{y} \\ &= \boldsymbol{\Lambda}^{-1}\boldsymbol{Z}^\top\boldsymbol{y} \end{aligned} \tag{3.4}$$

The variance-covariance matrix of $\hat{\boldsymbol{\theta}}$ has the form

$$\mathrm{Var}(\hat{\boldsymbol{\theta}}) = \sigma^2\boldsymbol{\Lambda}^{-1} = \sigma^2 \overbrace{\begin{pmatrix} \lambda_1^{-1} & 0 & \cdots & 0 \\ 0 & \lambda_2^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_p^{-1} \end{pmatrix}}^{\text{Diagonal matrix } \boldsymbol{\Lambda}^{-1}} \tag{3.5}$$

Then

$$\operatorname{tr}\operatorname{Var}(\hat{\boldsymbol{\theta}}) = \sigma^2 \sum_{j=1}^{p} \frac{1}{\lambda_j}.$$

Therefore, small eigenvalues inflate the total variance. One remedy is to make the transform $\lambda_j \mapsto \lambda_j + k$.
It can be shown that

$$\operatorname{tr}\operatorname{Var}(\hat{\boldsymbol{\beta}}(k)) = \sigma^2 \sum_{j=1}^{p} \frac{\lambda_j}{(\lambda_j + k)^2}. \tag{3.6}$$

Some more algebra plays: Recall $\boldsymbol{C} = \sum_j \lambda_j \boldsymbol{\gamma}\boldsymbol{\gamma}^\top$. Then, it is easy to see the following

- $\boldsymbol{C}^{-1} = \sum_j \lambda_j^{-1} \boldsymbol{\gamma}\boldsymbol{\gamma}^\top$
- $\boldsymbol{C} + k\boldsymbol{I}_p = \sum_j (k + \lambda_j) \boldsymbol{\gamma}\boldsymbol{\gamma}^\top$
- $\boldsymbol{C}(k)^{-1} = \sum_j (k + \lambda_j)^{-1} \boldsymbol{\gamma}\boldsymbol{\gamma}^\top$
- $\boldsymbol{C}(k)\boldsymbol{C}\boldsymbol{C}(k) = \sum_j \lambda_j (k + \lambda_j)^{-2} \boldsymbol{\gamma}\boldsymbol{\gamma}^\top$

Eigenvalue shrinkage

Remember we earlier termed k as the shrinkage parameter. What is the rationale? It can be shown that

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\Gamma}[\boldsymbol{\Lambda}^{-2}\boldsymbol{\Lambda}]\boldsymbol{\Gamma}^\top \boldsymbol{X}^\top \boldsymbol{y} \text{ and } \hat{\boldsymbol{\beta}}(k) = \boldsymbol{\Gamma}[(\boldsymbol{\Lambda}^2 + k\boldsymbol{I}_p)^{-1}\boldsymbol{\Lambda}]\boldsymbol{\Gamma}^\top \boldsymbol{X}^\top \boldsymbol{y} \tag{3.7}$$

The difference between the two estimators $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\beta}}(k)$ with respect to the spectral decomposition is in the square bracket terms (see Saleh et al., 2019 for the proof with SVD). Let

$$\boldsymbol{\Lambda}^{-1} = \begin{pmatrix} \lambda_1^{-1} & 0 & \cdots & 0 \\ 0 & \lambda_2^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_p^{-1} \end{pmatrix}, \ (\boldsymbol{\Lambda}^2 + k\boldsymbol{I}_p)^{-1}\boldsymbol{\Lambda} = \begin{pmatrix} \frac{\lambda_1}{\lambda_1^2+k} & 0 & \cdots & 0 \\ 0 & \frac{\lambda_2}{\lambda_2^2+k} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\lambda_p}{\lambda_p^2+k} \end{pmatrix}$$

Since $\frac{\lambda_j}{\lambda_j^2+k} \leq \frac{1}{\lambda_j}$ the ridge parameter shrinks the λ_js.

Orthonormal case

First, we have orthogonal and orthonormal!
For the orthonormal case, $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}_p$. Then

$$\hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \boldsymbol{y}, \quad \hat{\boldsymbol{\beta}}(k) = \frac{1}{1+k}\boldsymbol{X}^\top \boldsymbol{y}.$$

Therefore,

$$\hat{\boldsymbol{\beta}}(k) = \frac{1}{1+k}\hat{\boldsymbol{\beta}}. \tag{3.8}$$

Thus

$$\hat{\boldsymbol{\beta}}(k) \xrightarrow{k\to 0} \hat{\boldsymbol{\beta}}, \quad \hat{\boldsymbol{\beta}}(k) \xrightarrow{k\to\infty} 0$$

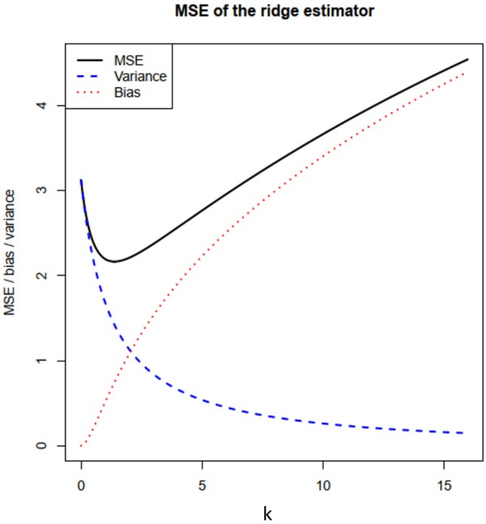
Recall $\mathrm{MSE}(\hat{\boldsymbol{\beta}}) = \mathrm{tr}\,\mathrm{Var}(\hat{\boldsymbol{\beta}}) + \mathrm{Bias}(\hat{\boldsymbol{\beta}})^\top \mathrm{Bias}(\hat{\boldsymbol{\beta}})$. Then, it can be shown

$$\begin{aligned} \mathrm{MSE}(\hat{\boldsymbol{\beta}}) &= p\sigma^2 \\ \mathrm{MSE}(\hat{\boldsymbol{\beta}}(k)) &= \frac{1}{(1+k)^2} p\sigma^2 + \frac{k^2}{(1+k)^2} \boldsymbol{\beta}^\top \boldsymbol{\beta}. \end{aligned} \tag{3.9}$$

The minimum of $\mathrm{MSE}(\hat{\boldsymbol{\beta}}(k))$ is achieved at

$$k = \frac{p\sigma^2}{\boldsymbol{\beta}^\top \boldsymbol{\beta}} \tag{3.10}$$

Trade-off


MSE of the ridge estimator
MSE
Variance
Bias
MSE / bias / variance
k
0
1
2
3
4
0
5
10
15

Lab2

```
library(glmnet)
n = 1000
p = 100
nzc = trunc(p/10)
x = matrix(rnorm(n * p), n, p)
beta = rnorm(nzc)
fx = x[, seq(nzc)]
eps = rnorm(n) * 5
y = drop(fx + eps)
cvob1 = cv.glmnet(x, y, alpha = 0)
plot(cvob1)
coef(cvob1)
```

We will come back to Remedy 3 at a later time in “penalized estimation”.

Assignments

Q1 : Show that centering the observations $(\boldsymbol{x}, y)$ eliminates the intercept parameter from the linear regression model.

Q2 : What is Tikhonov regularization? Why do you put constraint?

Q3 : Verify the equality (3.5).

Q4 : Prove equation (3.6).

Q5 : Write the R code for a high-dimensional case $p > n$ and make five predictions.

Thank you for your attention.